TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Development of an energy-efficient laser-based drying system for continuous electrode manufacturing of lithium-ion and solid-state batteries*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (x), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (x), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical - Electronics			
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	*Manufacturing and Measurement/Evaluation Equipment*		
Concept of Proposed Technology	The proposed technology focuses on the development of a laser-based drying system for continuous roll-to-roll (R2R) electrode manufacturing for lithium-ion and next-generation solid-state batteries. Conventional thermal drying processes used in electrode production are highly energy-intensive and represent a major bottleneck in battery manufacturing, particularly with increasing web speeds and thicker electrode designs. The proposed laser drying approach enables localized, selective, and dynamically controllable energy input directly into the wet electrode coating. The system is designed as a modular, drop-in laser drying unit that can be integrated into existing coating and calendering lines. By combining tailored optical energy distribution with real-time temperature monitoring and closed-loop process control, uniform solvent removal can			

	be achieved while minimizing thermal stress on active materials, binders, and current collectors. The technology is applicable to both anode and cathode production and is especially relevant for advanced electrode concepts, including high-loading electrodes and solid-state battery architectures.
Rationale for Joint R&D Project	The development of laser-based drying technology for battery electrodes requires close collaboration between laser technology experts and industrial partners involved in battery manufacturing equipment and electrode production. Fraunhofer ILT contributes strong expertise in laser-material interaction, optical system design, and process diagnostics, while Korean SMEs provide essential know-how in battery manufacturing processes, system integration, and industrial requirements. Joint R&D is necessary to: • adapt laser drying concepts to real industrial electrode materials and solvents, • validate process stability under continuous manufacturing conditions, • scale laboratory results toward pilot-line operation, and • ensure industrial relevance, cost-efficiency, and future commercialization. The collaboration enables accelerated technology transfer from laboratory-scale concepts to pilot-scale demonstrators aligned with the needs of the Korean battery industry.
Key Words	1. 1. Laser drying 2. Electrode manufacturing 3. Battery production process 4. Energy-efficient manufacturing 5. Solid-state batteries

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	The main objectives of this R&D project are: • Development of a laser-based drying system suitable for continuous roll-to-roll electrode manufacturing. • Design of an optical system enabling homogeneous and controllable energy distribution across the electrode width. • Implementation of a closed-loop temperature and process control system to ensure stable drying conditions. • Demonstration of efficient solvent removal for anode and cathode electrodes, including materials relevant for solid-state batteries. • Validation of the process at pilot-line scale to reach TRL 6.			
R&D Target Specifications	Energy consumption compared to conventional hot-air drying: **< 80%** Residual solvent content after laser drying: **< 1%** Web speed during drying: **> 5 m/min** Uniform temperature distribution across electrode width No degradation of electrochemical performance compared to reference electrodes			
R&D Plan	Phase 1 – Process analysis and system design (TRL 4) Analysis of electrode materials, solvents, and coating structures Modeling of laser–material interaction and heat transfer Design of optical system with multiple independently controllable heating zones Phase 2 – System development and control implementation Development of laser module and beam-shaping optics Integration of temperature sensors and real-time monitoring Implementation of closed-loop control algorithms Phase 3 – Pilot-line validation (TRL 5–6) Integration of laser drying module into a pilot R2R coating line Drying trials for anode, cathode, and solid-state relevant electrodes Evaluation of residual solvent content, adhesion, microstructure, and electrochemical performance Assessment of energy efficiency and scalability			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer ILT will: • develop and design the laser drying concept and optical system, • investigate laser–material interactions relevant to battery electrodes, • implement temperature monitoring and process control strategies, • perform material characterization (microstructure, adhesion, solvent content), • support pilot-line validation and process optimization, and • provide background IP for research purposes during the project period.
Expected R&R of Korean SME Partner	The Korean SME partner is expected to: • provide industrial requirements and manufacturing constraints, • supply electrode materials and representative test samples, • integrate the laser drying module into a demonstrator or pilot system, • perform system-level testing under industrial conditions, • lead commercialization planning, including IP strategy and market analysis, and • contribute to scale-up and future product development.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcomes of this joint R&D project include: • Development of a prototype-level laser drying system for battery electrode manufacturing. • Demonstration of high-throughput drying at relevant web speeds with reduced energy consumption. • Generation of new intellectual property jointly owned by project partners. • Strengthening of Korean battery manufacturing competitiveness through advanced process technology. • Creation of a foundation for follow-up projects targeting full commercialization (TRL 8–9). The technology contributes directly to energy-efficient manufacturing, carbon reduction in battery production, and next-generation battery technologies including solid-state batteries.

6. References

(1) Fraunhofer ILT internal research on laser-based thermal processing
(2) Prior collaborative projects and feasibility studies in battery manufacturing technologies

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization